

Excelerate
Data Visualization Associate Early Internship

SLU 0408 DVA Team 40

2nd Week Deliverables Report

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Week 2 – ETL Process of Master Table (Team Report)

Internship: Data Visualization Associate (Excelerate Virtual Internship)

Week: 2 – Master Table Creation, ETL Procedure & Data Quality Checks

Team Contribution: Joint report covering schema design, stored procedure development, and validation of the ETL process.

Introduction

During the first week of the Excelerate Data Visualization Internship, our team progressed from exploratory data analysis toward the integration and engineering phase. Then in 2nd week, the focus was on consolidating multiple raw datasets into a single structured Master Table, supported by an automated ETL procedure. This stage was crucial for ensuring data consistency, accuracy, and efficiency, laying a strong foundation for the advanced visualization and analytical work in the upcoming weeks.

Objectives

The objectives for this week included:

- Design and create the Master Table schema
- Define constraints, keys, and indexes to ensure performance and integrity
- Develop a stored procedure to automate extraction, transformation, and loading (ETL)
- Apply cleaning rules for duplicates, missing values, and inconsistent text formats
- Validate the process through quality checks and document findings in a Data Quality Report

List of Assigned Datasets

1. User_data.csv
2. Opportunity_Raw.csv
3. Cohort_Raw.csv
4. Marketing_data.csv
5. Learner_Opportunity_Raw.xlsx
6. Cognito_Raw2.xlsx

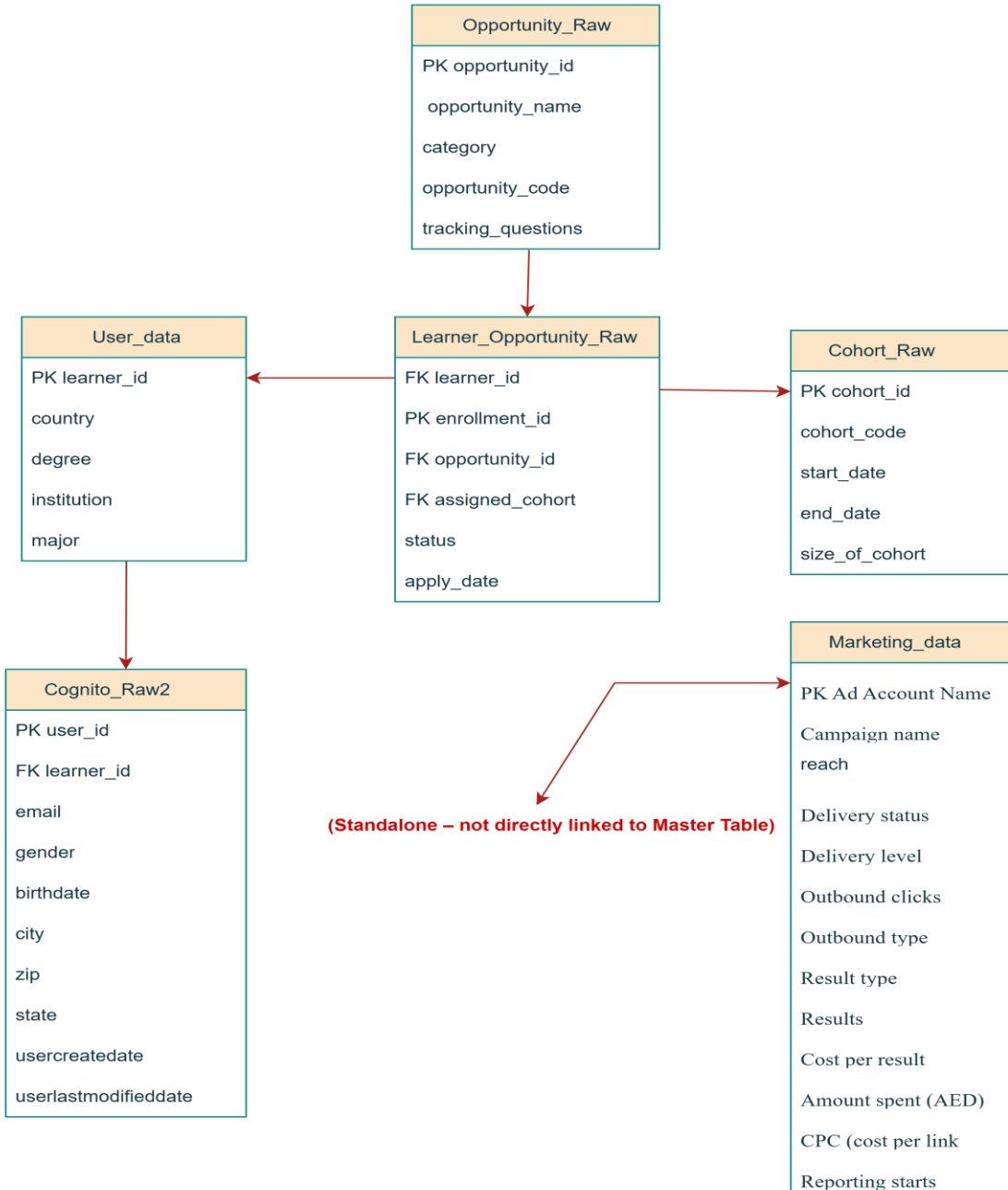
By the end of the week, the team successfully delivered a fully integrated Master Table, an ETL stored procedure that standardized and cleaned the raw inputs, and a Data Quality Report that documented issues detected, cleaning logic applied, and validation methods used. These deliverables ensure traceability, reproducibility, and high-quality data integration, establishing a solid groundwork for visualization and advanced analytics in the following stages of the internship.

Data Visualization Associate Early Internship

Exploring All Datasets:

Dataset Name	Column Name	Role in Schema	Key Type
User_data.csv	learner_id	Unique identifier for learners	PK
	country, degree, institution, major	Attributes linked to learner	—
Opportunity_Raw.csv	opportunity_id	Unique identifier for opportunities	PK
	opportunity_name, category, opportunity_code, tracking_questions	Attributes linked to opportunity	—
Cohort_Raw.csv	cohort_id	Unique identifier for cohorts	PK
	cohort_code	Links learners to cohorts	Alternate Key
Marketing_data.csv	campaign_name (or Ad Account Name)*	Identifiers for marketing campaigns (no direct link to Master Table)	PK (local only)
	Other campaign attributes (reach, clicks, cost, etc.)	Analytics fields	—
Learner_Opportunity_Raw.csv	enrollment_id	Unique enrollment record	PK
	learner_id	Connects to User_data.learner_id	FK → User_data(learner_id)
	assigned_cohort	Connects to Cohort_Raw.cohort_code	FK → Cohort_Raw(cohort_code)
	opportunity_id (from mapping)	Connects to Opportunity_Raw.opportunity_id	FK → Opportunity_Raw(opportunity_id)
	status, apply_date	Enrollment metadata	—
Cognito_Raw2.csv	user_id	Unique identifier for authentication records	PK
	learner_id (via mapping in Master Table) email, gender, birthdate, city, zip, state, usercreatedate, userlastmodifieddate	Links Cognito users with learners Attributes	FK → User_data(learner_id)

Schema Diagram:



Planning of Master Table:

Column Name	Source Dataset(s)	Reason for Inclusion
enrollment_id (PK)	Generated during ETL	A new, unique ID created for each row; ensures that every application by a learner to an opportunity is uniquely identified for accurate analysis and management.
learner_id (FK)	Learner_Raw, Cognito_Raw2	Unique student identifier; connects demographic info (Cognito) with academic background (Learner) and actions (LearnerOpportunity).
opportunity_id (FK)	Opportunity_Raw, LearnerOpportunity_Raw	Identifies the specific opportunity a learner applied for; links to details like name and category to analyze popularity and relevance.
cohort_id (FK)	Cohort_Raw, LearnerOpportunity_Raw	Identifies the cohort associated with an application; connects learners to timelines with cohort start/end dates.
email	Cognito_Raw2	Provides essential contact info and acts as a human-readable learner identifier.
gender	Cognito_Raw2	Basic demographic info for segmentation and trend analysis across genders.
birthdate	Cognito_Raw2	Enables age-based analysis, such as age distribution of applicants.
city, state, country	Cognito_Raw2, Learner_Raw	Key geographic data for analyzing regional trends, target markets, and distribution of learners.
degree, institution, major	Learner_Raw	Academic background details; useful for analyzing correlations between education and opportunity success.
opportunity_name, opportunity_category	Opportunity_Raw	Adds descriptive context to opportunity_id; improves readability of reports and analysis.
cohort_start_date, cohort_end_date	Cohort_Raw	Provides program timeline; essential for planning, tracking, and analyzing program duration.
apply_date	LearnerOpportunity_Raw	Exact timestamp of application; critical for time-series analysis and tracking trends.
status	LearnerOpportunity_Raw	Shows application outcome (e.g., Accepted, Rejected, In Progress); key metric for program and learner success.

Keys and Relationships: The Table's Skeleton

- **Primary Key (PK): `enrollment_id`**
 - It's a unique ID for every single row in the master table.
 - It's essential because a learner can apply multiple times; this key distinguishes each unique application.
 - It guarantees data integrity and prevents confusion.
- **Foreign Keys (FKs): `learner_id`, `opportunity_id`, `cohort_id`**
 - These keys act as the "glue" that links data back to the original source tables.
 - They enforce a critical rule: you can't have an application for a learner or opportunity that doesn't exist.
 - This prevents "orphan records" and ensures all data is valid and connected.

Overall Reasoning for This Design

- **Creates a Single Source of Truth:** Combines five fragmented files into one complete narrative for each student.
- **Enables Complex Analysis:** Allows you to easily answer detailed questions that cross multiple datasets (e.g., "What is the academic background of applicants for X opportunity?").
- **Provides a Robust & Scalable Structure:** The design is strong, can grow with more data, and is perfectly optimized for building dashboards and reports.

Master Table Creation: Query:

```

55 -- Create the final Master_Table by selecting from transformed and joined data
56 CREATE TABLE Master_Table AS
57 WITH Cognito_Clean AS (
58     -- Clean user data: convert dates and handle NULLs
59     SELECT
60         user_id,
61         email,
62         NULLIF(gender, 'NULL') AS gender,
63         TO_TIMESTAMP(NULLIF(NULLIF(UserCreateDate, 'NULL'), ''), 'YYYY-MM-DD"T"HH24:MI:SS.MS"Z") AS user_create_date,
64         TO_TIMESTAMP(NULLIF(NULLIF(UserLastModifiedDate, 'NULL'), ''), 'YYYY-MM-DD"T"HH24:MI:SS.MS"Z") AS user_last_modified_date,
65         TO_DATE(NULLIF(NULLIF(birthdate, 'NULL'), ''), 'MM/DD/YYYY') AS birthdate,
66         NULLIF(NULLIF(city, 'NULL'), '') AS city,
67         NULLIF(NULLIF(state, 'NULL'), '') AS state
68     FROM Cognito_Raw
69 ),
70 Learner_Clean AS (
71     -- Clean learner academic data: remove ID prefix
72     SELECT
73         REPLACE(learner_id, 'Learner#', '') AS learner_id,
74         NULLIF(NULLIF(country, 'NULL'), '') AS country,
75         NULLIF(NULLIF(degree, 'NULL'), '') AS degree,
76         NULLIF(NULLIF(institution, 'NULL'), '') AS institution,
77         NULLIF(NULLIF(major, 'NULL'), '') AS major
78     FROM Learner_Raw
79 ),
80 Opportunity_Clean AS (
81     -- Clean opportunity data: remove ID prefix
82     SELECT
83         REPLACE(opportunity_id, 'Opportunity#', '') AS opportunity_id,

```

```

Query Query History
79 ),
80 Opportunity_Clean AS (
81     -- Clean opportunity data: remove ID prefix
82     SELECT
83         REPLACE(opportunity_id, 'Opportunity#', '') AS opportunity_id,
84         opportunity_name,
85         category,
86         opportunity_code
87     FROM Opportunity_Raw
88 ),
89 Cohort_Clean AS (
90     -- Clean cohort data: convert Unix timestamps (handle scientific notation)
91     SELECT
92         cohort_code,
93         TO_TIMESTAMP(CAST(start_date AS DOUBLE PRECISION) / 1000) AS start_date,
94         TO_TIMESTAMP(CAST(end_date AS DOUBLE PRECISION) / 1000) AS end_date,
95         size::integer AS size -- convert size from VARCHAR to integer
96     FROM Cohort_Raw
97     WHERE start_date <> 'NULL' AND end_date <> 'NULL'
98 ),
99 LearnerOpportunity_Clean AS (
100     -- Clean enrollment data: remove prefixes, map status, and convert date
101     SELECT
102         REPLACE(learner_id, 'Learner#', '') AS learner_id,
103         REPLACE(opportunity_id, 'Opportunity#', '') AS opportunity_id,
104         NULLIF(NULLIF(assigned_cohort, 'NULL'), '') AS assigned_cohort,
105         TO_TIMESTAMP(NULLIF(NULLIF(apply_date, 'NULL'), ''), 'YYYY-MM-DD"T"HH24:MI:SS.MS"Z") AS apply_date,
106         CASE
107             WHEN status = '1030' THEN 'Applied'
108             WHEN status = '1070' THEN 'Enrolled'
109             WHEN status = '1120' THEN 'Completed'
110             WHEN status = '1130' THEN 'Withdrawn'
111             ELSE 'Unknown'
112         END AS status
113     FROM LearnerOpportunity_Raw
114 )
115
116 -- Final SELECT to create the master table
117 SELECT
118     lo.learner_id,
119     lo.opportunity_id,
120     lo.assigned_cohort,
121     lo.apply_date,
122     lo.status,
123     l.country,
124     l.degree,
125     l.institution,
126     l.major,
127     c.email,
128     c.gender,
129     c.user_create_date,
130     c.user_last_modified_date,
131     c.birthdate,
132     c.city,
133     c.state,
134     o.opportunity_name,
135     o.category AS opportunity_category,

```

Activate Windows
Go to Settings to activate Windows.

```

Query Query History
107     WHEN status = '1030' THEN 'Applied'
108     WHEN status = '1070' THEN 'Enrolled'
109     WHEN status = '1120' THEN 'Completed'
110     WHEN status = '1130' THEN 'Withdrawn'
111     ELSE 'Unknown'
112 END AS status
113 FROM LearnerOpportunity_Raw
114 )
115
116 -- Final SELECT to create the master table
117 SELECT
118     lo.learner_id,
119     lo.opportunity_id,
120     lo.assigned_cohort,
121     lo.apply_date,
122     lo.status,
123     l.country,
124     l.degree,
125     l.institution,
126     l.major,
127     c.email,
128     c.gender,
129     c.user_create_date,
130     c.user_last_modified_date,
131     c.birthdate,
132     c.city,
133     c.state,
134     o.opportunity_name,
135     o.category AS opportunity_category,

```

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```

Query Query History
122     lo.status,
123     l.country,
124     l.degree,
125     l.institution,
126     l.major,
127     c.email,
128     c.gender,
129     c.user_create_date,
130     c.user_last_modified_date,
131     c.birthdate,
132     c.city,
133     c.state,
134     o.opportunity_name,
135     o.category AS opportunity_category,
136     o.opportunity_code,
137     co.start_date AS cohort_start_date,
138     co.end_date AS cohort_end_date,
139     co.size AS cohort_size
140 FROM LearnerOpportunity_Clean lo
141 LEFT JOIN Learner_Clean l
142     ON lo.learner_id = l.learner_id
143 LEFT JOIN Cognito_Clean c
144     ON lo.learner_id = c.user_id
145 LEFT JOIN Opportunity_Clean o
146     ON lo.opportunity_id = o.opportunity_id
147 LEFT JOIN Cohort_Clean co
148     ON lo.assigned_cohort = co.cohort_code;
149
150

```

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Output:

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	learner_id text	opportunity_id text	assigned_cohort text	apply_date timestamp with time zone	status text	country text	degree text
1	00004f18-8b86-4fe4-ad7e-6c8d988f5335	000000000GWQACX5C452MH...	B43E1MB	2023-07-23 08:13:40.371+06	Enrolled	Nigeria	Undergradu
2	00010567-1336-433c-a941-a612b3d2fb...	00000000100JPM3A0WJ85T68...	BEXFE80	2024-11-17 21:36:30.848+06	Enrolled	Kenya	Graduate St
3	0001ca2c-7bec-4a33-833c-b844a29f4d...	000000000GN2A0AY7XK8C5FZPP	BWAG78I	2024-11-12 22:30:13.77+06	Completed	Nigeria	Graduate St
4	0001ca2c-7bec-4a33-833c-b844a29f4d...	0000000010GG17ZMAP4TWQVB...	BB8ZJ2M	2024-11-12 22:09:46.267+06	Enrolled	Nigeria	Graduate St
5	0001ca2c-7bec-4a33-833c-b844a29f4d...	0000000010BDV2YMK8R1RN7K...	BC6601N	2024-11-25 18:13:18.925+06	Enrolled	Nigeria	Graduate St
6	0003bed9-d9d9-49a7-a755-a9562aaa0d...	0000000010SAZXDAE05AN2CG...	B6R9JVJ	2025-02-12 04:48:31.821+06	Enrolled	Pakistan	Graduate St
7	0003dac8-45a1-4dd9-9c9b-9c3cfc7060cf	0000000010SAZXDAE05AN2CG...	[null]	2025-02-21 15:06:46.315+06	Unknown	India	Undergradu
8	0004295c-717e-4953-b3bf-fff2daf0e903	0000000010RQJXA9NNKDEW5R...	BRG4BZF	2024-04-11 22:32:58.407+06	Enrolled	Nigeria	Undergradu
9	00049a81-94a9-4b25-92ed-62d017f3b6...	0000000010GG17ZMAP4TWQVB...	BRHGPWW	2024-04-01 10:44:29.898+06	Enrolled	Philippines	Undergradu
10	00084381-3917-4d03-9639-0a501aa68c...	0000000010WCBS50CYGDY97E...	BGRQZ2N	2025-01-30 18:44:44.217+06	Enrolled	India	High Schoo
11	000926e9-66af-4e40-a788-538e74b9a0...	0000000010AWJ1XABSV8Y81F...	BYDCIHI	2025-01-23 08:20:23.487+06	Enrolled	Pakistan	Graduate St
12	0009e8ae-95a0-4b58-b5dd-06b90d14ea...	000000000GHB4N83QX9KJM48...	BG9O2UX	2025-01-19 05:57:59.903+06	Enrolled	Nepal	Undergradu
13	000bce06-dbee-4d9f-988c-26ed26ab99...	0000000010ZKNWGJ45840TGR...	BGVPUY2	2023-04-24 10:04:05.347+06	Enrolled	India	Undergradu
14	000e0279-b2a9-4891-ad00-0284d52a06...	000000000GHB4N83QX9KJM48...	BUUPEAO	2024-09-29 21:47:46.85+06	Enrolled	Pakistan	Undergradu
15	000e3bb4-383d-4412-a5a2-e807661793...	000000000GNTFT74MZT893VC...	B570DHJ	2024-05-22 10:42:45.383+06	Enrolled	Egypt	Undergradu
16	000ff2e3-c653-4153-bd88-3bd4ae7f1c2a	0000000010GG17ZMAP4TWQVB...	BKCDFG7	2024-03-28 02:55:17.906+06	Completed	Nigeria	Undergradu
17	00108b20-0435-4ad7-9d27-8878f635de...	000000000GWQACX5C452MH...	BGFN3JO	2024-05-16 20:16:33.721+06	Enrolled	India	Undergradu
18	0012caf1-83ee-4a25-9f78-71d6fcb18ee6	0000000010CD3CAK3FZK9VXKR...	BAPXRMZ	2024-06-25 05:35:46.801+06	Unknown	India	Undergradu
19	0012caf1-83ee-4a25-9f78-71d6fcb18ee6	0000000010YM7EFAPFPZPDZVE9	BGHT2K9	2024-06-12 12:52:41.243+06	Completed	India	Undergradu
20	00000000000000000000000000000000	00000000000000000000000000000000	00000000	2024-06-05 05:06:46.70+06	Enrolled	India	Undergradu

Total rows: 113602

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	degree text	institution text	major text
1	Undergraduate Student	Federal University of Technology Owerri	Civil Engineering
2	Graduate Student	UNICAF UNIVERSITY	Environmental Sustainability
3	Graduate Student	Nasarawa State University, Keffi	Accounting
4	Graduate Student	Nasarawa State University, Keffi	Accounting
5	Graduate Student	Nasarawa State University, Keffi	Accounting
6	Graduate Student	CTTI College KP Campus	Part Time
7	Undergraduate Student	Allenhouse institute of technology	Electronics and Communication
8	Undergraduate Student	Caleb University	Computer Science
9	Undergraduate Student	Our lady of fatima university	Nursing
10	High School Student	Saket International School	Medicine
11	Graduate Student	Air University	Marketing
12	Undergraduate Student	Global college international	Travel and Tourism
13	Undergraduate Student	MG University	Cyber Security
14	Undergraduate Student	FAST NUCES, Islamabad	Computer Science
15	Undergraduate Student	Self Hitham Nabill salah	Undergraduate
16	Undergraduate Student	University of Lagos	Philosophy
17	Undergraduate Student	ABES Engineering College	Computer Engineering
18	Undergraduate Student	Baikunthi Devi Kanya Mahavidyalaya	Business Administration
19	Undergraduate Student	Baikunthi Devi Kanya Mahavidyalaya	Business Administration
20	Undergraduate Student	Baikunthi Devi Kanya Mahavidyalaya	Business Administration

Total rows: 113602

Query complete 00:00:00.596

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	opportunity_name character varying (255)	opportunity_category character varying (100)	opportunity_code character varying (100)	cohort_start_date timestamp with time zone	cohort_end_date timestamp with time zone	cohort_size integer
1	Data Visualization Early Internship	Internship	I352815	2023-08-21 09:53:20+06	2023-09-26 09:46:40+06	1
2	Jump Start: Developing your Emotional Intelligence	Course	URFOJWP	2024-06-18 06:20:00+06	2025-06-18 04:40:00+06	1
3	Career Essentials: Getting Started with Your Professional Journey	Course	U533812	2024-10-18 06:06:40+06	2025-10-18 23:53:20+06	1
4	Data Visualization Associate Early Internship	Internship	I4ZTKD3	2024-12-02 06:40:00+06	2024-12-31 13:26:40+06	[truncated]
5	Accountant	Career	AJENF5Z	2024-11-25 21:53:20+06	2024-12-10 23:00:00+06	
6	Project Management Associate Early Internship	Internship	IP5EYAL	2025-03-03 06:00:00+06	2025-04-01 12:46:40+06	[truncated]
7	Project Management Associate Early Internship	Internship	IP5EYAL	[null]	[null]	
8	UX Redesign Challenge	Competition	MEXA1GG	2024-02-16 00:00:00+06	2024-05-07 00:26:40+06	1
9	Data Visualization Associate Early Internship	Internship	I4ZTKD3	2024-04-15 06:13:20+06	2024-05-14 13:00:00+06	1
10	CPR/AED Certification	Course	USNZAI6	2024-05-28 04:46:40+06	2026-02-10 05:33:20+06	
11	Business Development Virtual Internship	Internship	I2KYO99	2025-02-10 07:13:20+06	2025-03-11 14:00:00+06	
12	Project Management Early Internship	Internship	I448557	2025-02-03 05:46:40+06	2025-03-04 12:33:20+06	
13	Cybersecurity: Defensive Hacking	Internship	I710882	2023-06-05 07:53:20+06	2023-07-03 08:06:40+06	
14	Project Management Early Internship	Internship	I448557	2024-10-07 06:13:20+06	2024-11-07 12:13:20+06	
15	Digital Marketing Early Internship	Internship	I476315	2024-07-01 05:26:40+06	2024-07-30 12:13:20+06	
16	Data Visualization Associate Early Internship	Internship	I4ZTKD3	2024-07-01 05:26:40+06	2024-07-30 12:13:20+06	
17	Data Visualization Early Internship	Internship	I352815	2024-09-02 07:20:00+06	2024-10-01 11:20:00+06	
18	Digital Strategy Early Internship	Internship	I25190A	2024-08-05 07:06:40+06	2024-09-04 12:06:40+06	
19	Design Dynamics: Journey into Creative Solutions	Event	EVX291M	2024-06-13 20:46:40+06	2024-06-30 23:53:20+06	1
20	Project Management Early Internship	Internship	I448557	2024-06-06 07:06:40+06	2024-06-26 07:06:40+06	
21	Project Management Early Internship	Internship	I448557	2024-06-06 07:06:40+06	2024-06-26 07:06:40+06	

Total rows: 113602

Query complete 00:00:00.596

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Data Output Messages Notifications						
Showing rows: 1 to 1000 Page No: 1 of 114						
	ortunity_name racter varying (255)	opportunity_category character varying (100)	opportunity_code character varying (100)	cohort_start_date timestamp with time zone	cohort_end_date timestamp with time zone	cohort_size integer
1	a Visualization Early Internship	Internship	I352815	2023-08-21 09:53:20+06	2023-09-26 09:46:40+06	1000
2	np Start: Developing your Emotional Intelligence	Course	URFOJWP	2024-06-18 06:20:00+06	2025-06-18 04:40:00+06	1600
3	eer Essentials: Getting Started with Your Professional Journey	Course	U533812	2024-10-18 06:06:40+06	2025-10-18 23:53:20+06	1580
4	a Visualization Associate Early Internship	Internship	I4ZTKD3	2024-12-02 06:40:00+06	2024-12-31 13:26:40+06	800
5	countant	Career	AJENF5Z	2024-11-25 21:53:20+06	2024-12-10 23:00:00+06	500
6	ject Management Associate Early Internship	Internship	IP5EYAL	2025-03-03 06:00:00+06	2025-04-01 12:46:40+06	800
7	ject Management Associate Early Internship	Internship	IP5EYAL	[null]	[null]	[null]
8	Redesign Challenge	Competition	MEXA1GG	2024-02-16 00:00:00+06	2024-05-07 00:26:40+06	1000
9	a Visualization Associate Early Internship	Internship	I4ZTKD3	2024-04-15 06:13:20+06	2024-05-14 13:00:00+06	500
10	R/AED Certification	Course	USNZAI6	2024-05-28 04:46:40+06	2026-02-10 05:33:20+06	1780
11	usiness Development Virtual Internship	Internship	I2KYO99	2025-02-10 07:13:20+06	2025-03-11 14:00:00+06	800
12	ject Management Early Internship	Internship	I448557	2025-02-03 05:46:40+06	2025-03-04 12:33:20+06	800
13	persecurity: Defensive Hacking	Internship	I710882	2023-06-05 07:53:20+06	2023-07-03 08:06:40+06	300
14	ject Management Early Internship	Internship	I448557	2024-10-07 06:13:20+06	2024-11-07 12:13:20+06	800
15	ital Marketing Early Internship	Internship	I476315	2024-07-01 05:26:40+06	2024-07-30 12:13:20+06	800
16	a Visualization Associate Early Internship	Internship	I4ZTKD3	2024-07-01 05:26:40+06	2024-07-30 12:13:20+06	800
17	a Visualization Early Internship	Internship	I352815	2024-09-02 07:20:00+06	2024-10-01 11:20:00+06	800
18	ital Strategy Early Internship	Internship	I25190A	2024-08-05 07:06:40+06	2024-09-04 12:06:40+06	800
19	sign Dynamics: Journey into Creative Solutions	Event	EVX291M	2024-06-13 20:46:40+06	2024-06-30 23:53:20+06	1500
20	ject Management Early Internship	Internship	I448557	2024-06-05 07:06:40+06	2024-06-05 07:06:40+06	800
Total rows: 113602 Query complete 00:00:00.596 CRLF Ln 2, Col 1						

ETL Process (Extract, Transform & Load) Of Master Table:

Data Cleaning :

- Correction of datatypes in each column in master_table:

Query:

```

4 -- ETL Process of Master_Table
5
6 -- Datatypes of all column
7 SELECT
8     column_name,
9     data_type
10 FROM information_schema.columns
11 WHERE table_name = 'master_table'
12 ORDER BY ordinal_position;
13

```

Output:

Data Output Messages Notifications		
column_name	data_type	
1 learner_id	text	
2 opportunity_id	text	
3 assigned_cohort	text	
4 apply_date	timestamp with time zone	
5 status	text	
6 country	text	
7 degree	text	
8 institution	text	
9 major	text	
10 email	character varying	
11 gender	text	
12 user_create_date	timestamp with time zone	
13 user_last_modified_date	timestamp with time zone	
14 birthdate	date	
15 city	text	
16 state	text	
17 opportunity_name	character varying	
18 opportunity_category	character varying	
19 opportunity_code	character varying	
20 cohort_start_date	timestamp with time zone	
21 cohort_end_date	timestamp with time zone	
22 cohort_size	integer	
Total rows: 22 Query complete 00:00:00.158		

- Checking if there is actual null/missing values in each column:
Query: (for checking null counts):

```

Query Query History
18 -- Checking Missing/null values
19 SELECT 'learner_id' AS column_name, COUNT(*) - COUNT(learner_id) AS null_count FROM Master_Table
20 UNION ALL
21 SELECT 'opportunity_id', COUNT(*) - COUNT(opportunity_id) FROM Master_Table
22 UNION ALL
23 SELECT 'assigned_cohort', COUNT(*) - COUNT(assigned_cohort) FROM Master_Table
24 UNION ALL
25 SELECT 'apply_date', COUNT(*) - COUNT(apply_date) FROM Master_Table
26 UNION ALL
27 SELECT 'status', COUNT(*) - COUNT(status) FROM Master_Table
28 UNION ALL
29 SELECT 'country', COUNT(*) - COUNT(country) FROM Master_Table
30 UNION ALL
31 SELECT 'degree', COUNT(*) - COUNT(degree) FROM Master_Table
32 UNION ALL
33 SELECT 'institution', COUNT(*) - COUNT(institution) FROM Master_Table
34 UNION ALL
35 SELECT 'major', COUNT(*) - COUNT(major) FROM Master_Table
36 UNION ALL
37 SELECT 'email', COUNT(*) - COUNT(email) FROM Master_Table
38 UNION ALL
39 SELECT 'gender', COUNT(*) - COUNT(gender) FROM Master_Table
40 UNION ALL
41 SELECT 'user_create_date', COUNT(*) - COUNT(user_create_date) FROM Master_Table
42 UNION ALL
43 SELECT 'user_last_modified_date', COUNT(*) - COUNT(user_last_modified_date) FROM Master_Table
44 UNION ALL
45 SELECT 'birthdate', COUNT(*) - COUNT(birthdate) FROM Master_Table
46 UNION ALL
47 SELECT 'city', COUNT(*) - COUNT(city) FROM Master_Table
48 UNION ALL
49 SELECT 'state', COUNT(*) - COUNT(state) FROM Master_Table
50 UNION ALL
51 SELECT 'opportunity_name', COUNT(*) - COUNT(opportunity_name) FROM Master_Table
52 UNION ALL
53 SELECT 'opportunity_category', COUNT(*) - COUNT(opportunity_category) FROM Master_Table
54 UNION ALL
55 SELECT 'opportunity_code', COUNT(*) - COUNT(opportunity_code) FROM Master_Table
56 UNION ALL
57 SELECT 'cohort_start_date', COUNT(*) - COUNT(cohort_start_date) FROM Master_Table
58 UNION ALL
59 SELECT 'cohort_end_date', COUNT(*) - COUNT(cohort_end_date) FROM Master_Table
60 UNION ALL
61 SELECT 'cohort_size', COUNT(*) - COUNT(cohort_size) FROM Master_Table;
62

```

Activ:
Go to S

Output:

Before

Data Output			Messages	Notifications
	column_name	null_count		
	text	bigint		
1	learner_id	0		
2	cohort_size	13318		
3	cohort_end_date	13318		
4	cohort_start_date	13318		
5	opportunity_code	0		
6	opportunity_category	0		
7	opportunity_name	0		
8	opportunity_id	0		
9	apply_date	188		
10	status	0		
11	degree	593		

12	major	596
13	gender	600
14	user_last_modified_date	275
15	city	600
16	assigned_cohort	13318
17	country	223
18	institution	593
19	email	275
20	user_create_date	275
21	birthdate	600
22	state	601
Total rows: 22		Query complete 00:00:00.468



VS After (cleaning)

Query: (for removing nulls & replacing valid values)

```
Query Query History
64 -- fixing null values in each column
65 -- Replace NULLs in text / character columns with 'Unknown'
66 v UPDATE Master_Table
67 SET
68     email = CASE WHEN email IS NULL THEN 'Unknown' ELSE email END,
69     gender = CASE WHEN gender IS NULL THEN 'Unknown' ELSE gender END,
70     city = CASE WHEN city IS NULL THEN 'Unknown' ELSE city END,
71     state = CASE WHEN state IS NULL THEN 'Unknown' ELSE state END,
72     country = CASE WHEN country IS NULL THEN 'Unknown' ELSE country END,
73     degree = CASE WHEN degree IS NULL THEN 'Unknown' ELSE degree END,
74     institution = CASE WHEN institution IS NULL THEN 'Unknown' ELSE institution END,
75     major = CASE WHEN major IS NULL THEN 'Unknown' ELSE major END,
76     opportunity_name = CASE WHEN opportunity_name IS NULL THEN 'Unknown' ELSE opportunity_name END,
77     opportunity_category = CASE WHEN opportunity_category IS NULL THEN 'Unknown' ELSE opportunity_category END,
78     opportunity_code = CASE WHEN opportunity_code IS NULL THEN 'Unknown' ELSE opportunity_code END,
79     assigned_cohort = CASE WHEN assigned_cohort IS NULL THEN 'Unknown' ELSE assigned_cohort END,
80     status = CASE WHEN status IS NULL THEN 'Unknown' ELSE status END;
81
82 -- Replace NULLs in integer columns with 0
83 v UPDATE Master_Table
84 SET
85     cohort_size = CASE WHEN cohort_size IS NULL THEN 0 ELSE cohort_size END;
86
87
88 -- Replace NULLs in date / timestamp columns with current date / timestamp
88 v UPDATE Master_Table
89 SET
90     user_create_date = CASE WHEN user_create_date IS NULL THEN CURRENT_TIMESTAMP ELSE user_create_date END,
91     user_last_modified_date = CASE WHEN user_last_modified_date IS NULL THEN CURRENT_TIMESTAMP ELSE user_last_modified_date END,
92     birthdate = CASE WHEN birthdate IS NULL THEN CURRENT_DATE ELSE birthdate END,
93     apply_date = CASE WHEN apply_date IS NULL THEN CURRENT_TIMESTAMP ELSE apply_date END,
94     cohort_start_date = CASE WHEN cohort_start_date IS NULL THEN CURRENT_TIMESTAMP ELSE cohort_start_date END,
95     cohort_end_date = CASE WHEN cohort_end_date IS NULL THEN CURRENT_TIMESTAMP ELSE cohort_end_date END; Activate Windows
```

Output:

Data Output			Messages	Notifications
	column_name	null_count		
	text	bigint		
1	learner_id	0		
2	cohort_size	0		
3	cohort_end_date	0		
4	cohort_start_date	0		
5	opportunity_code	0		
6	opportunity_category	0		
7	opportunity_name	0		
8	state	0		
9	city	0		
10	birthdate	0		
11	opportunity_id	0		
12	apply_date	0		
13	degree	0		
14	institution	0		
15	email	0		
16	user_create_date	0		
17	assigned_cohort	0		
18	status	0		
19	country	0		
20	major	0		
21	gender	0		
22	user_last_modified_date	0		
Total rows: 22			Query complete 00:00:00.461	



- Checking if there is text type “NULL” values in each column:

Query: (for checking NULL text type values) :

```
Query  Query History
98  --checked NULL text type values
99  ✓ SELECT 'email' AS column_name, COUNT(*) AS null_text_count FROM Master_Table WHERE email = 'NULL'
100 UNION ALL
101 SELECT 'gender', COUNT(*) FROM Master_Table WHERE gender = 'NULL'
102 UNION ALL
103 SELECT 'city', COUNT(*) FROM Master_Table WHERE city = 'NULL'
104 UNION ALL
105 SELECT 'state', COUNT(*) FROM Master_Table WHERE state = 'NULL'
106 UNION ALL
107 SELECT 'country', COUNT(*) FROM Master_Table WHERE country = 'NULL'
108 UNION ALL
109 SELECT 'degree', COUNT(*) FROM Master_Table WHERE degree = 'NULL'
110 UNION ALL
111 SELECT 'institution', COUNT(*) FROM Master_Table WHERE institution = 'NULL'
112 UNION ALL
113 SELECT 'major', COUNT(*) FROM Master_Table WHERE major = 'NULL'
114 UNION ALL
115 SELECT 'opportunity_name', COUNT(*) FROM Master_Table WHERE opportunity_name = 'NULL'
116 UNION ALL
117 SELECT 'opportunity_category', COUNT(*) FROM Master_Table WHERE opportunity_category = 'NULL'
118 UNION ALL
119 SELECT 'opportunity_code', COUNT(*) FROM Master_Table WHERE opportunity_code = 'NULL'
120 UNION ALL
121 SELECT 'assigned_cohort', COUNT(*) FROM Master_Table WHERE assigned_cohort = 'NULL'
122 UNION ALL
123 SELECT 'status', COUNT(*) FROM Master_Table WHERE status = 'NULL';
124
```

Output:

	column_name	null_text_count
	text	bigint
1	status	0
2	opportunity_category	0
3	degree	0
4	gender	0
5	email	0
6	institution	0
7	city	0
8	assigned_cohort	0
9	opportunity_name	0
10	state	0
11	opportunity_code	0
12	major	0
13	country	0



- Checked if there is any duplicate records and Total Records:

Query:

```
149
150
151 -- 1. Record Count Validation
152 v SELECT 'Source Enrollments' AS table_name, COUNT(*) AS record_count FROM LearnerOpportunity_Raw
153 UNION ALL
154 SELECT 'Master Table' AS table_name, COUNT(*) AS record_count FROM Master_Table;
155
```

Output:

Data Output

Messages

Notifications

SQL

	table_name	record_count
	text	bigint
1	Source Enrollments	113602
2	Master Table	113602

118

SELECT COUNT(*) AS total_records

119

FROM public.Master_Table;

120

Data Output

Messages

Notifications

SQL

	total_records
	bigint
1	113602

SQL

As Source Enrollments Records =Master table Records, there is no duplicate records in Master_Table.

Checking & Removing URLs (Don't want to specify) values from “gender” column:

Query:

```
125
126 -- counting URLs values (Don't want to specify) in gender column
127 v SELECT COUNT(*) AS count_URL_gender
128 FROM Master_Table
129 WHERE gender = 'Don't want to specify';
130
131 -- removing URLs values, replacing unknown
132 v UPDATE Master_Table
133 SET gender = 'Unknown'
134 WHERE gender = 'Don't want to specify';
135
```

Before VS After (cleaning)

Data Output

Messages

Notifications

SQL

	count_url_gender
	bigint
1	366

Data Output

Messages

Notifications

SQL

	count_url_gender
	bigint
1	0

SQL

Final Master Table After Cleaned (Viewing first 15 rows):

```
1 -- Final Master_Table After cleaning
2 ✓ SELECT * FROM public.master_table
3 LIMIT 15
```

Data Output

Messages

Notifications

SQL

Showing rows: 1 to 15

Page No: 1

of 1

	learner_id text	opportunity_id text	assigned_cohort text	apply_date timestamp with time zone	status text	country text	degree text	institution text
1	719f5def-3d27-4192-bd9e-d9acece7b7a	0000000010CD3CAK3FZK9VXX...	Unknown	2024-04-02 09:25:40.762+06	Applied	Nigeria	Not in Education	Federal University of Tech
2	719f9f90-3704-4f2e-bdc9-9438ac1b51a0	0000000010GG17ZMAP4TWQV...	BCXC21V	2024-10-22 14:08:39.915+06	Enrolled	Kenya	Graduate Student	Dedan Kimathi University
3	719f9e12-c2de-41f5-826a-f5cde2c97a99	0000000123GB9ZSVRTC3YS9T716N	BRTCV69	2023-05-26 22:45:31.208+06	Enrolled	India	Undergraduate Student	New Horizon College of E
4	7fa0847b-d88d-4d08-8f0d-2a6d16396b1f	0000000010SAZXDAE05AN2CG...	Unknown	2024-09-20 12:53:33.134+06	Applied	Nigeria	Graduate Student	University Of Port Harcou
5	7fa0847b-d88d-4d08-8f0d-2a6d16396b1f	000000000GN2A0AY7XK8CSFZ...	BSEV9QO	2024-09-19 09:29:58.652+06	Enrolled	Nigeria	Graduate Student	University Of Port Harcou
6	7fa0847b-d88d-4d08-8f0d-2a6d16396b1f	00000000010ZWEVXFZSARHJSE...	BE7X8PZ	2024-11-29 23:16:09.268+06	Enrolled	Nigeria	Graduate Student	University Of Port Harcou
7	7fa0847b-d88d-4d08-8f0d-2a6d16396b1f	00000000010EY9NM8M6HJ12D6S...	BP9ZV19	2025-01-30 18:16:54.339+06	Enrolled	Nigeria	Graduate Student	University Of Port Harcou
8	7fa0847b-d88d-4d08-8f0d-2a6d16396b1f	000000000GH84N83QX9KJM48...	BYZZO85	2024-09-19 22:55:39.376+06	Enrolled	Nigeria	Graduate Student	University Of Port Harcou
9	7fa0e13b-bc2c-4d76-95dd-9b740b25f03e	0000000000GWQAXCSX45C2MH...	BPPMQ44	2023-07-21 19:48:57.946+06	Enrolled	Nigeria	Not in Education	Obafemi Awolowo Univer
10	7fa1023d-b035-45ad-bf50-07d38ca5fe3f	0000000000GWQAXCSX45C2MH...	BGBWW5I	2024-05-13 22:30:24.532+06	Enrolled	Nigeria	Undergraduate Student	Editing state university
11	7fa1023d-b035-45ad-bf50-07d38ca5fe3f	00000000010GG17ZMAP4TWQV...	BLHWY32	2024-06-06 18:01:35.372+06	Enrolled	Nigeria	Undergraduate Student	Editing state university
12	7fa1efc3-9529-4e79-8d65-aac1f69fa21a	0000000000GWQAXCSX45C2MH...	BPPMQ44	2023-07-07 12:15:40.108+06	Enrolled	Ghana	Not in Education	Valley View University
13	7fa24cff-e6f6-4d67-8e12-64d4cd1fd8a	00000000010VWCXG6F542KJ9...	BYV7FAG	2025-01-15 22:11:06.333+06	Enrolled	Bangladesh	Graduate Student	University of Dhaka
14	7fa33d9e-16c5-45f2-9c71-21706b31e01d	00000000010BRJY9TVC6XG9...	BZHWCB	2024-12-17 06:08:05.928+06	Enrolled	United States	Undergraduate Student	De La Salle University Ma
15	7fa33d9e-16c5-45f2-9c71-21706b31e01d	00000000010ANNE7X7T1SS2JJPK	BSS2JV0	2024-11-14 00:06:46.262+06	Unkno...	United States	Undergraduate Student	De La Salle University Ma

Total rows: 15

Query complete 00:00:00.296

CRLF

Ln 141, Col 1

Data Output

Messages

Notifications

SQL

Showing rows: 1 to 15

Page No: 1 of 1

	institution text	major text	email character varying (255)	gender text	user_create_date timestamp with time zone	user_last_modified_date timestamp with time zone	birthdate date	city text
1	Federal University of Technology, Minna	Biochemistry	nwaforqueen12345@gmail.com	Female	2024-04-02 08:41:55.984+06	2024-09-27 16:13:27.161+06	1998-03-09	Abuja
2	Dedan Kimathi University of Technolo...	Nutrition and Dietetics	wambuwiaweru026@gmail.com	Female	2024-10-22 13:06:20.326+06	2024-11-04 12:53:12.968+06	1998-03-14	Nairo
3	New Horizon College of Engineering	Computer Science and Engineering	vijaykumar2209100125@gmail.com	Male	2023-05-26 22:34:31.682+06	2023-08-29 12:22:02.907+06	2002-03-22	Visak
4	University Of Port Harcourt	Communication and Media Studies	stephanie.dapper1@gmail.com	Female	2024-09-19 08:04:57.073+06	2024-09-27 14:05:29.951+06	2001-10-29	Port H
5	University Of Port Harcourt	Communication and Media Studies	stephanie.dapper1@gmail.com	Female	2024-09-19 08:04:57.073+06	2024-09-27 14:05:29.951+06	2001-10-29	Port H
6	University Of Port Harcourt	Communication and Media Studies	stephanie.dapper1@gmail.com	Female	2024-09-19 08:04:57.073+06	2024-09-27 14:05:29.951+06	2001-10-29	Port H
7	University Of Port Harcourt	Communication and Media Studies	stephanie.dapper1@gmail.com	Female	2024-09-19 08:04:57.073+06	2024-09-27 14:05:29.951+06	2001-10-29	Port H
8	University Of Port Harcourt	Communication and Media Studies	stephanie.dapper1@gmail.com	Female	2024-09-19 08:04:57.073+06	2024-09-27 14:05:29.951+06	2001-10-29	Port H
9	Obafemi Awolowo University	Data Analytics	modupeade89@gmail.com	Female	2023-07-21 19:33:37.904+06	2024-09-27 16:13:27.383+06	1989-06-18	Abuja
10	Editing state university	Bsc	iloriyomide88@gmail.com	Male	2024-05-13 22:15:39.349+06	2024-05-13 22:22:54.667+06	2003-05-18	Ibada
11	Editing state university	Bsc	iloriyomide88@gmail.com	Male	2024-05-13 22:15:39.349+06	2024-05-13 22:22:54.667+06	2003-05-18	Ibada
12	Valley View University	Other	adolphinekorba@yahoo.com	Female	2023-07-07 11:36:37.486+06	2024-09-27 16:13:27.605+06	1997-05-07	Accra
13	University of Dhaka	Disaster Management	mayishafahmida111@gmail.com	Female	2025-01-15 21:59:16.108+06	2025-01-15 22:10:04.703+06	2002-08-11	Jheni
14	De La Salle University Manila	Computer Engineering	z4nd3riv0@gmail.com	Male	2024-11-13 23:15:35.062+06	2025-01-13 10:33:21.061+06	2005-09-30	Manil
15	De La Salle University Manila	Computer Engineering	z4nd3riv0@gmail.com	Male	2024-11-13 23:15:35.062+06	2025-01-13 10:33:21.061+06	2005-09-30	Manil

Total rows: 15

Query complete 00:00:00.296

CRLF

Ln 141, Col 1

Ln 141, Col 1



Summary of Data Quality After ETL process of Master Table:

Issues Detected	Cleaning Logic Applied	Testing Methodology
Datatype inconsistencies	Converted all columns to correct datatype: • TEXT for IDs and textual fields • INTEGER for numeric fields (cohort_size) • DATE for birthdate, cohort_start_date, cohort_end_date • TIMESTAMP WITH TIME ZONE for apply_date, user_create_date, user_last_modified_date	Verified schema using Master_Table; tested date arithmetic and numeric operations; spot-checked ID columns remain as TEXT
Missing / NULL values in TEXT columns	Replaced literal "NULL" text and SQL NULLs with 'Unknown'	Counted NULL / 'NULL' values before and after cleaning for each column; confirmed replacements
Missing / NULL values in numeric columns	Replaced NULLs with 0	Counted NULL values before and after; verified totals remain consistent
Missing / NULL values in DATE / TIMESTAMP columns	Replaced NULLs with current date / timestamp	Counted NULL values before and after; spot-checked sample records to confirm updated timestamps
Gender column containing placeholder text	Replaced entries like "Don't want to specify" with 'Unknown'	Spot-checked gender column; counted occurrences before and after replacement
Null / missing join keys	Ensured join keys exist for relationships	Tested joins with Learner_Raw, Cognito_Raw, Opportunity_Raw, Cohort_Raw; confirmed no missing matches

Conclusion:

In Week 2, the team focused on creating the Master Table and implementing the ETL process. All raw datasets were integrated into a single structured table, ensuring proper relationships through primary and foreign keys. The ETL procedure standardized text fields, handled NULLs and placeholder values, removed duplicates, and converted dates and status codes into consistent formats. These steps ensured that the Master Table is clean, reliable, and ready for analysis. The resulting Master Table serves as a robust, standardized dataset, forming the backbone for all subsequent data visualization, reporting, and analytical tasks, with no remaining inconsistencies, missing values, or unreliable entries.